



Studienjahr <i>Academic Year</i>	2023/2024		
Studiengruppe(n) <i>Study Group(s)</i>	BA, IM, LM		
Prüfung <i>Exam</i>	Klausur / <i>Written examination</i>		
Modul(e) <i>Module(s)</i>	Principles of Finance		
Zugelassene Hilfsmittel <i>Permitted aids</i>	Einsprachiges Wörterbuch, nicht-programmierbarer Taschenrechner / <i>monolingual dictionary, non-programmable calculator</i>		
Seiten Aufgabenstellung / Anlagen <i>Pages tasks / appendices</i>	14 Seiten inkl. Deckblatt / <i>14 pages incl. front page</i>	Formulary (3 pages)	
Prüfungsdatum / -zeit / -dauer <i>Examination date/time/duration</i>	4.10.2024	8 – 9.30 a.m.	90 Minuten <i>Minutes</i>
Prüfer <i>Examiner</i>	Hr. Prof. Prigge (modulverantwortlich)		

Bitte leserlich ausfüllen! *Please complete legibly!*

Matrikelnummer <i>Matriculation number</i>			
Studiengruppe <i>Study Group</i>			
Anzahl zusätzlicher Seiten Papier <i>No. of additional sheets of paper</i>			
Klausurbeginn /-ende <i>Examination start / end</i>			

Punkteverteilung *Scores*

Aufgabe <i>Task</i>	1	2	3	4	Σ
Maximal <i>Maximum</i>	35	20	20	25	100
Erreicht <i>Achieved</i>					

Note <i>Grade:</i>	Prüfer <i>Examiner:</i>	Unterschrift Prüfer <i>Signature of the examiner:</i>

Punkte-Noten-Schema *Grading Scheme*

Punkte <i>Score</i>	100-95	94-90	89-85	84-80	79-75	74-70	69-65	64-60	59-55	54-50	49-0
Note <i>Grade</i>	1,0	1,3	1,7	2,0	2,3	2,7	3,0	3,3	3,7	4,0	n.b.

a) Capital Budgeting

Year	1	2	3	4	5
Cash Flow	11.000.000	15.000.000	8.000.000	4.000.000	1.000.000



- a2) The higher education department is not the only department at Super AI Tec AG. The other departments also developed promising projects. However, as there is a shortage of AI engineers, it might be that not all projects with a positive NPV can be realized. The Chief Investment Officer of Super AI Tec AG compiled a list of all potential projects for next year, displaying their NPV and their need of AI engineer capacity (as headcount of AI engineers). What is the optimal investment program of Super AI Tec AG for next year, provided that they have a capacity of 80 AI engineers? Show your calculations. **(9 points)**

Project	NPV (EUR)	Required AI Engineer Capacity
AI Grading	result from a1)	20
P1	2.500.000,00	15
P2	3.500.000,00	18
P3	6.000.000,00	25
P4	6.500.000,00	35
P5	7.800.000,00	43
P6	3.300.000,00	17



b) WACC

For Super AI Tec AG, the following data is given:

- cost of equity 17%
- cost of debt 10%
- market value of equity EUR250 million
- market value of debt EUR200 million
- corporate tax rate 35%

b1) Please calculate Super AI Tec AG's cost of debt after taxes.
(3 points)

A large rectangular grid of small dots, intended for the student to write their calculation for the cost of debt after taxes.

b2) Please calculate Super AI Tec AG's company WACC. **(6 points)**

A large rectangular grid of small dots, intended for the student to write their calculation for the company WACC.



- b3) The company WACC you just calculated differs from the WACC applied in task a1). Why might this make sense? How can the difference be interpreted? **(5 points)**

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Aufgabe Task 2 Bonds

(20 Punkte Points)

Suppose that today you buy an 8% annual coupon bond for EUR1,150. The bond has 10 years to maturity and a face value of EUR1,000.

- a) If you hold the bond until maturity, what rate of return do you expect to earn on your investment (no approximation via interpolation required)?
(4 points)

- b) Two years from now, the yield to maturity (YTM) on your bond has declined by one percentage point and you decide to sell. What price will your bond sell for? [Hint: If you did not calculate the initial YTM under a), assume the initial YTM two years ago was 7%.] **(4 points)**



- c) Assuming you sold the bond at the price you calculated under b), what is the internal rate of return on your investment over your holding period of two years? Approximation via linear approximation is required. Please calculate two NPVs, one positive and one negative, using the interest rates of 8% and 9%. **(6 points)**

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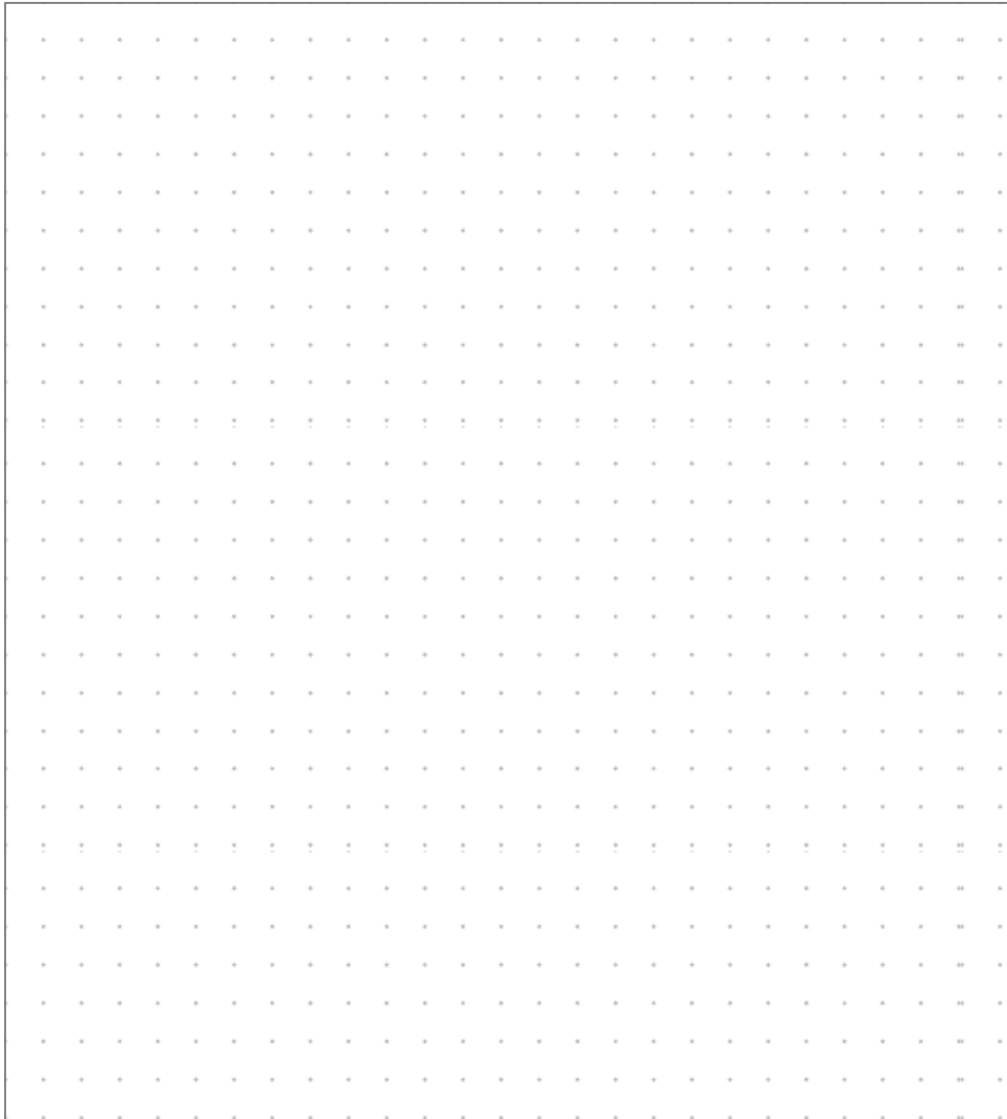
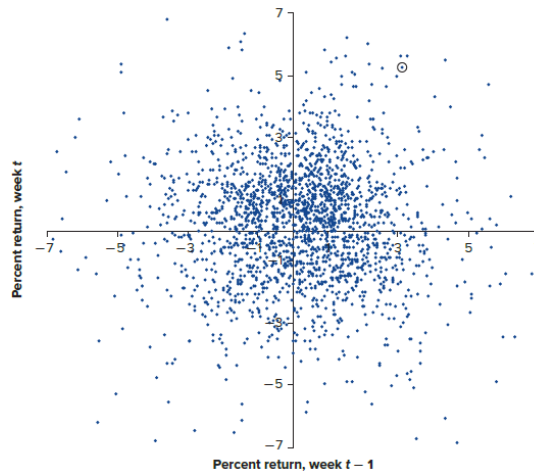
- d) Compare this return c) to the rate of return a) when you first bought the bond. Why are they different? [If you did not calculate a return under c) and not calculated one under a), then use a return for c) of 10% and a return for a) of 7%.] **(2 points)**

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- A large rectangular area filled with a uniform grid of small dots, intended for drawing a picture.

(20 Punkte Points)

- b) Please explain briefly the depicted chart below. Which form of market efficiency can be described with it? In how far can you outperform the market in that form of efficiency? **(6 points)**





Aufgabe Task 4 Risk and Return

(25 Punkte Points)

- a) Please explain the differences and commonalities between the risk profile of a single stock with a $\beta=1$ and a perfectly diversified portfolio with a $\beta=1$. (5 points)

A large rectangular area with a light gray dot grid pattern, intended for writing the answer to the question.

- | | Recession | Normal | Boom | Expected Return | Standard Deviation |
|---------|------------------|---------------|-------------|------------------------|---------------------------|
| Stock 1 | -9.00% | 8.00% | 21.00% | 6.67% | 12.28% |
| Stock 2 | -0.07% | 5.00% | 15.00% | 4.33% | 8.99% |
| Stock 3 | 10.00% | 4.00% | -5.00% | 3.00% | 6.16% |

-
- A large grid of dots for drawing a picture. The grid consists of 20 rows and 30 columns of dots. The first row has 28 dots, followed by rows 2 through 10 which each have 30 dots. Rows 11 through 19 each have 28 dots, and the final row (row 20) has 30 dots. This layout provides a space for a drawing that is approximately 10 cm wide and 15 cm high.



- b3) Please calculate the standard deviation of the portfolio after the second stock has been added. **(6 points)**

A large rectangular grid of small dots, intended for the student to show their calculations for the standard deviation of the portfolio.

- c) Applying the CAPM, what is the expected return for a stock with a $\beta = 1.2$ when the risk-free rate is at 3% and the market risk premium amounts to 7%? **(5 points)**

A large rectangular grid of small dots, intended for the student to show their calculations for the expected return using the CAPM.

Zusätzlicher Antwortbereich. Achten Sie bitte auf eine klare Zuordnung bzw. Nummerierung Ihrer Antworten. Additional answer area. Please make sure that your answers are clearly assigned or numbered.



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Formulary – Appendix to Exam Chapter 1

Future Value of a Cash Flow: $FV_n = C \times (1 + r) \times (1 + r) \times \dots \times (1 + r) = C \times (1 + r)^n$

Present Value of a Cash Flow: $PV = C \div (1 + r)^n = \frac{C}{(1 + r)^n}$

Net Present Value (NPV): compares the present value of cash inflows (benefits) to the present value of cash outflows (costs).

Present Value of a Perpetuity (first payment at time 1): $PV(C \text{ in perpetuity}) = \frac{C}{r}$

Present Value of an Annuity: $PV(\text{annuity of } C \text{ for } N \text{ periods with interest rate } r) = \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right)$

Future Value of an Annuity:

$$\begin{aligned} FV(\text{annuity}) &= PV \times (1 + r)^N \\ &= \frac{C}{r} \left(1 - \frac{1}{(1 + r)^N}\right) \times (1 + r)^N \\ &= C \times \frac{1}{r} ((1 + r)^N - 1) \end{aligned}$$

Present Value of a Growing Perpetuity: $PV(\text{growing perpetuity}) = \frac{C}{r - g}$

Present Value of a Growing Annuity: $PV = C \times \frac{1}{(r - g)} \left(1 - \left(\frac{1 + g}{(1 + r)}\right)^N\right)$

Approximate Internal Rate of Return (IRR) via Interpolation
(with $NPV_1 > 0$ and $NPV_2 < 0$): $r \approx r_1 - \frac{NPV_1}{NPV_2 - NPV_1} \cdot (r_2 - r_1)$

Chapter 2

Profitability Index: $\text{Profitability Index} = \frac{\text{Value Created}}{\text{Resource Consumed}} = \frac{NPV}{\text{Resource Consumed}}$

Chapter 3

Coupon Payment: $CPN = \frac{\text{Coupon Rate} \times \text{Face Value}}{\text{Number of Coupon Payments per Year}}$

Price of a Zero-Coupon Bond (FV=face value): $P = \frac{FV}{(1 + YTM)^n}$

Yield to Maturity of an n-Year Zero-Coupon Bond: $YTM = \left(\frac{FV}{P}\right)^{\frac{1}{n}} - 1$

Yield to Maturity of a Coupon Bond: discount rate y , that equates the PV of the bond's future cash flows with its price (=IRR): $P = CPN \times \frac{1}{YTM} \left(1 - \frac{1}{(1 + YTM)^N}\right) + \frac{FV}{(1 + YTM)^N}$

Chapter 4

Dividend-discount model (1 year): $P_0 = \left(\frac{\text{Div}_1 + P_1}{1 + r_E} \right)$

Total return, capital gain, dividend yield: $r_E = \frac{\text{Div}_1 + P_1}{P_0} - 1 = \underbrace{\frac{\text{Div}_1}{P_0}}_{\text{Dividend Yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{Capital Gain Rate}}$

Dividend-discount model (2 years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2 + P_2}{(1 + r_E)^2}$

Dividend-discount model (n years): $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$

$$P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \frac{\text{Div}_3}{(1 + r_E)^3} + \dots = \sum_{n=1}^{\infty} \frac{\text{Div}_n}{(1 + r_E)^n}$$

DDM, constant growth: $P_0 = \frac{\text{Div}_1}{r_E - g} \quad r_E = \frac{\text{Div}_1}{P_0} + g$

DDM, changing growth rates: $P_0 = \frac{\text{Div}_1}{1 + r_E} + \frac{\text{Div}_2}{(1 + r_E)^2} + \dots + \frac{\text{Div}_N}{(1 + r_E)^N} + \frac{1}{(1 + r_E)^N} \left(\frac{\text{Div}_{N+1}}{r_E - g} \right)$

Change in Earnings = New Investment × Return on New Investment

New Investment = Earnings × Retention Rate

Earnings Growth Rate = $\frac{\text{Change in Earnings}}{\text{Earnings}} = \text{Retention Rate} \times \text{Return on New Investment}$

$g = \text{Retention Rate} \times \text{Return on New Investment}$

Dividend: $\text{Div}_t = \underbrace{\frac{\text{Earnings}_t}{\text{Shares Outstanding}_t}}_{\text{EPS}_t} \times \text{Dividend Payout Rate}_t$

Chapter 5

Expected (Mean) Return: Calculated as a weighted average of the possible returns, where the weights correspond to the probabilities.

$$\text{Expected Return} = E[R] = \sum_R P_R \times R$$

Variance: The expected squared deviation from the mean.

$$\text{Var}(R) = E[(R - E[R])^2] = \sum_R P_R \times (R - E[R])^2$$

Variance Estimate Using Realized Returns:

$$\text{Var}(R) = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R})^2$$

Standard Deviation: The square root of the variance.

$$SD(R) = \sqrt{\text{Var}(R)}$$

$$\begin{aligned} \text{Portfolio rate of return} &= (\text{fraction of portfolio in asset 1} \times \text{rate of return on asset 1}) \\ &+ (\text{fraction of portfolio in asset 2} \times \text{rate of return on asset 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in asset } t \times \text{rate of return on asset } t) \end{aligned}$$

$$\begin{aligned} \text{Portfolio beta} &= (\text{fraction of portfolio in stock 1} \times \beta \text{ of stock 1}) \\ &+ (\text{fraction of portfolio in stock 2} \times \beta \text{ of stock 2}) \\ &\dots \\ &+ (\text{fraction of portfolio in stock } t \times \beta \text{ of stock } t) \end{aligned}$$

$$\text{Risk premium on the market} = r_m - r_f$$

$$\text{Risk premium on any asset } i = r_i - r_f$$

$$\text{SML equation: } r = r_f + \beta(r_m - r_f)$$

Chapter 6

$$r_{\text{assets}} = \frac{\text{total income}}{\text{value of investments}} = \frac{(D \times r_{\text{debt}}) + (E \times r_{\text{equity}})}{V}$$

$$r_{\text{assets}} = \frac{D}{V} \times r_{\text{debt}} + \frac{E}{V} \times r_{\text{equity}}$$

$$\text{After-tax cost of debt} = r_{\text{debt}} \times (1 - T_c).$$

$$\text{WACC} = \left[\frac{D}{V} \times (1 - T_c) r_{\text{debt}} \right] + \left[\frac{E}{V} \times r_{\text{equity}} \right]$$